

ECSE-4040

Digital Electronics

Lecture 6

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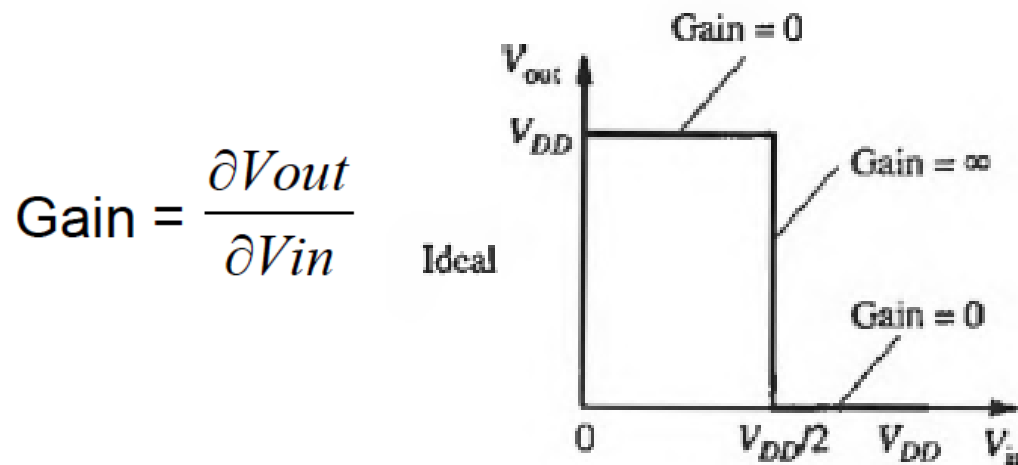
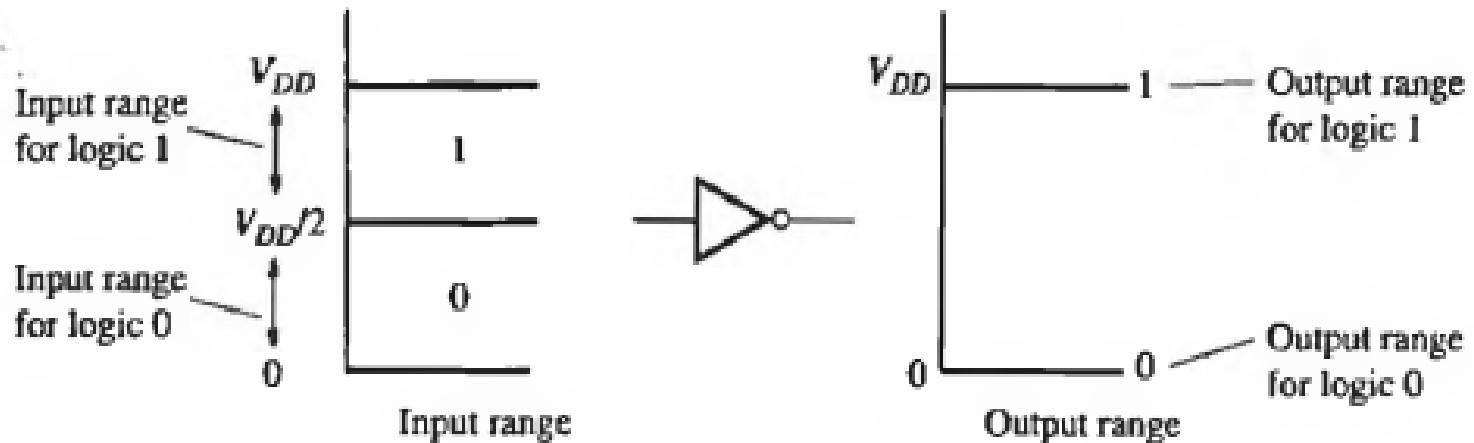
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Lecture Outline

- Ideal and Practical VTC
- Noise Margin
- Resistive-Load Inverter Design
- NMOS Inverter Design
 - Saturated Enhancement Load Design
 - Linear Enhancement Load Design
 - Depletion Load Design

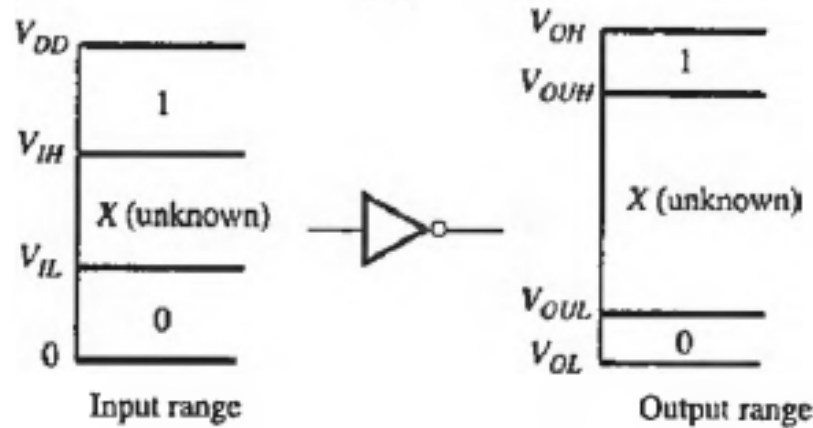
Ideal Inverter VTC



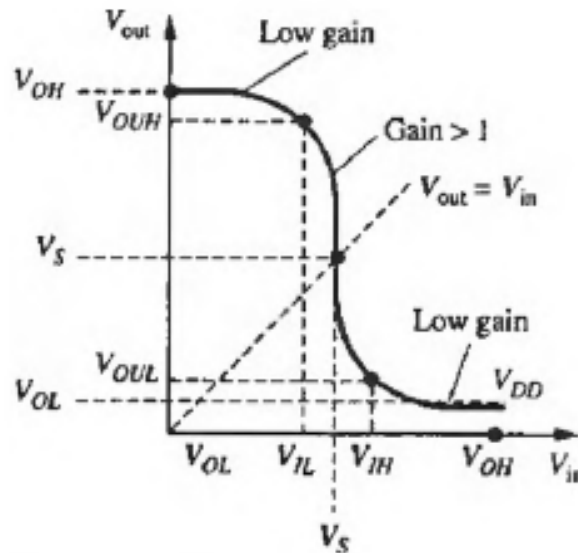
$$\text{Gain} = \frac{\partial V_{out}}{\partial V_{in}}$$

Ideal

Practical Inverter VTC

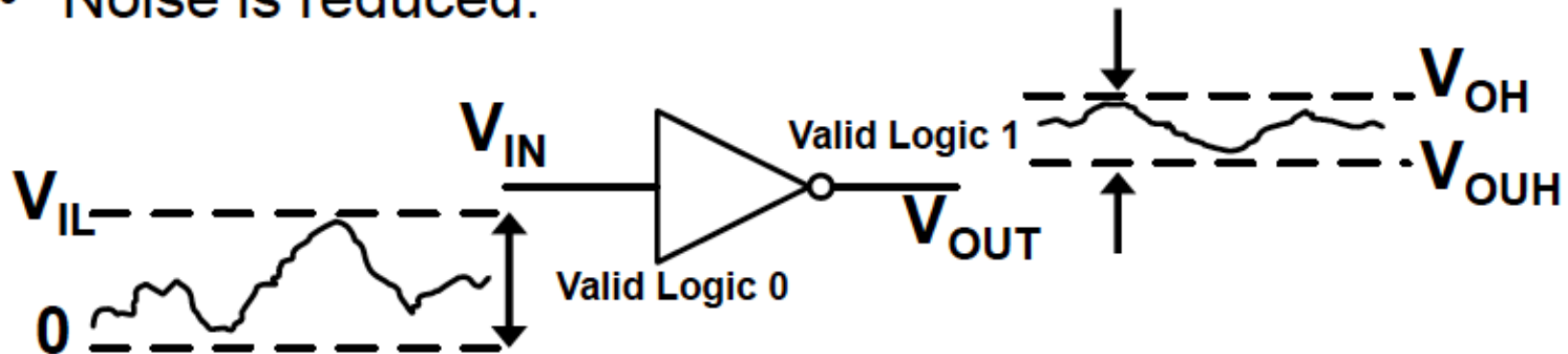


- Gain is -1 at V_{IL} and V_{IH}
- Important points:
 - V_{OH}, V_{OL}
 - V_S
 - V_{IL}, V_{IH}
 - V_{OUH}, V_{OUL}



Noise Attenuation

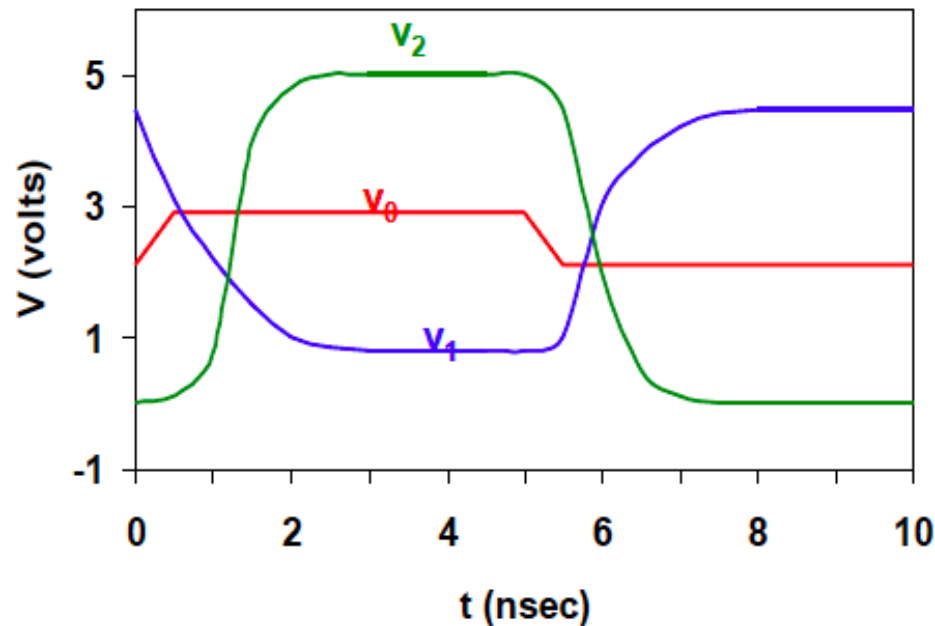
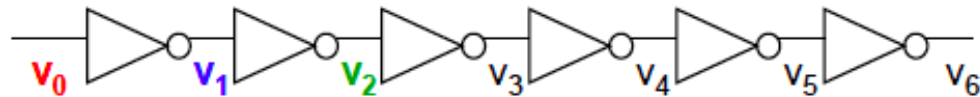
- What does this type of VTC imply about a noisy signal as it passes through a logic gate?
- Noise is reduced:



- Next, consider metrics for noise tolerance and the implications on the characteristics of the gate

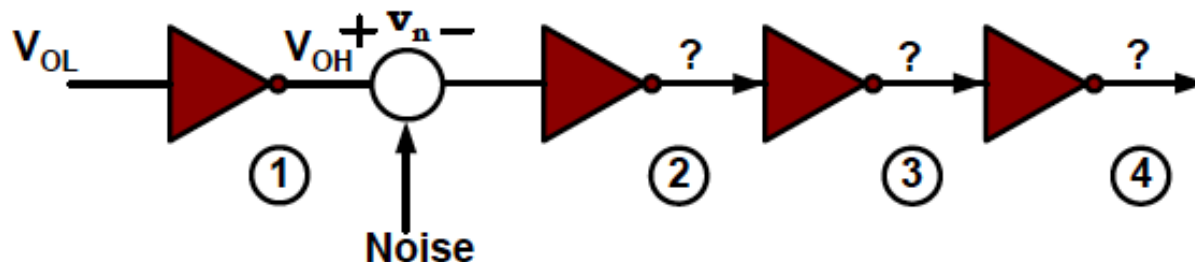
Regenerative Property

- A gate with regenerative property ensure that a disturbed signal converges back to a nominal voltage level



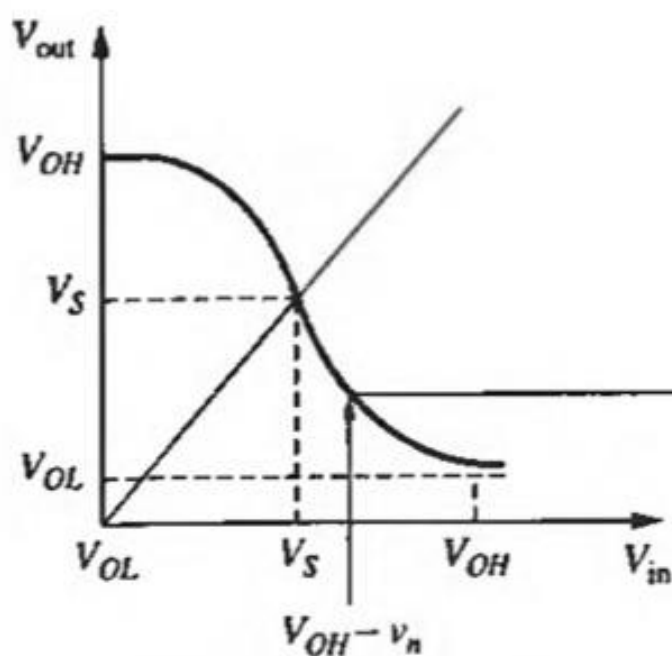
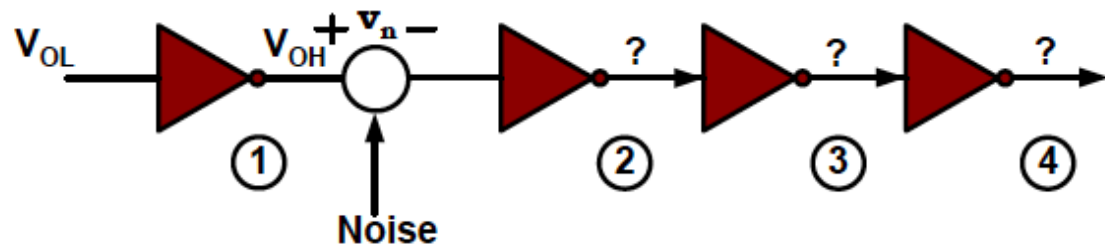
Single Source Noise Margin (SSNM)

- Simplest type of noise margin is the single-stage noise margin
- Defined as maximum noise, v_n , in a single stage that still allows subsequent stages to recover to the right value (regenerative property)

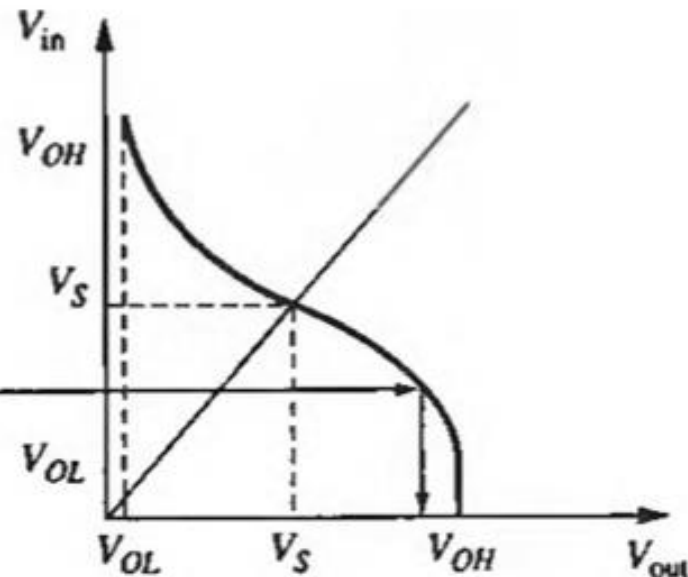


- In the above circuit $V_{i2} = V_{o1} - v_n = V_{OH} - v_n$
- For noise added to a high level input, the correct levels will be maintained if $V_{OH} - v_n > V_{TH}$
- For noise added to a low level input, correct levels will be maintained if $V_{OL} + v_n < V_{TH}$

Single Source Noise Margin (SSNM)

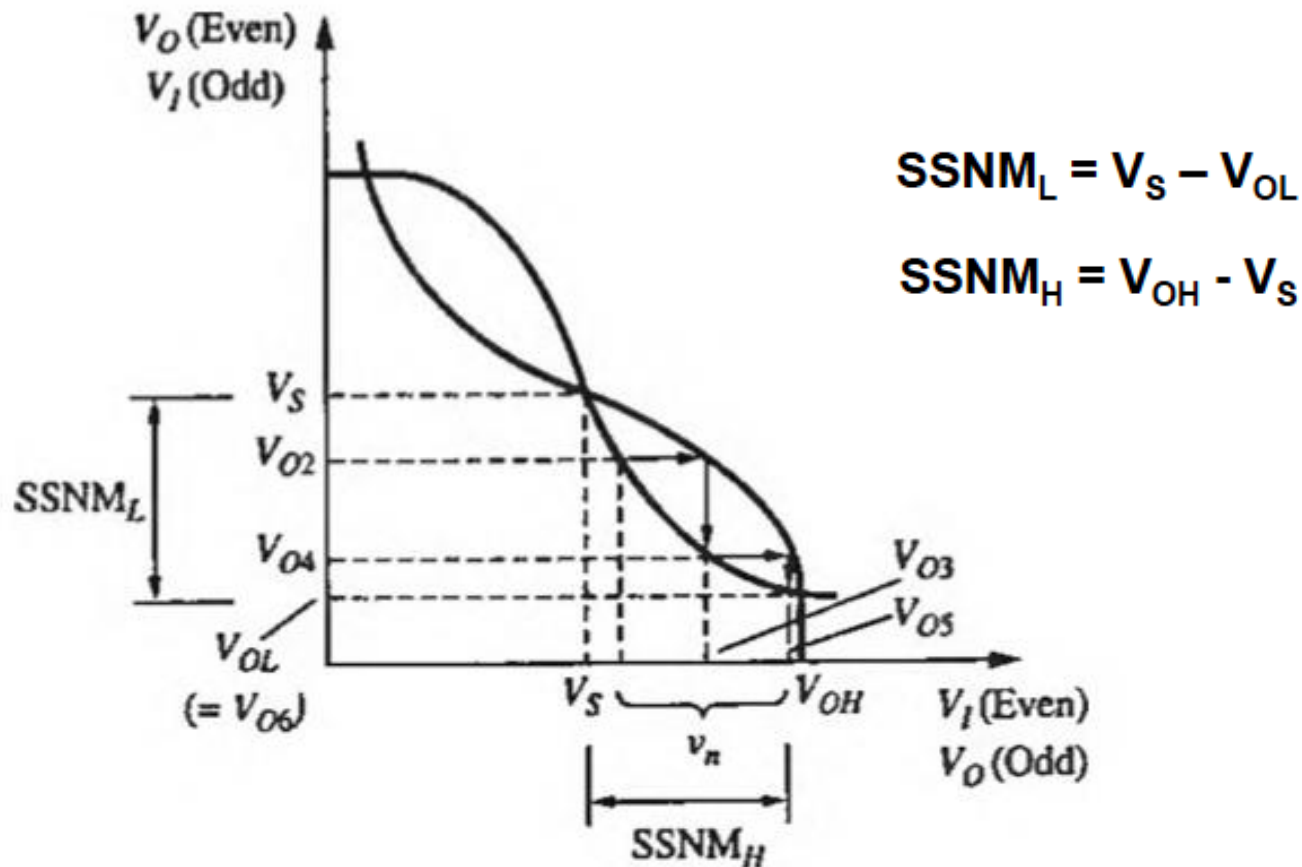


(a) Second inverter



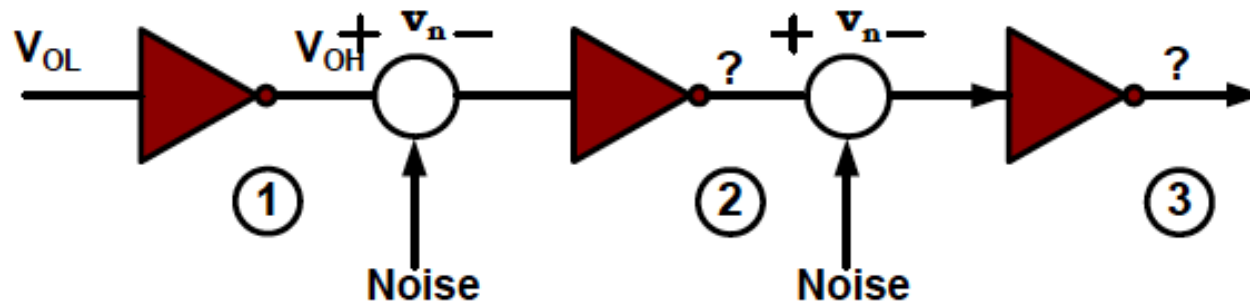
(b) Third inverter

Single Source Noise Margin (SSNM)



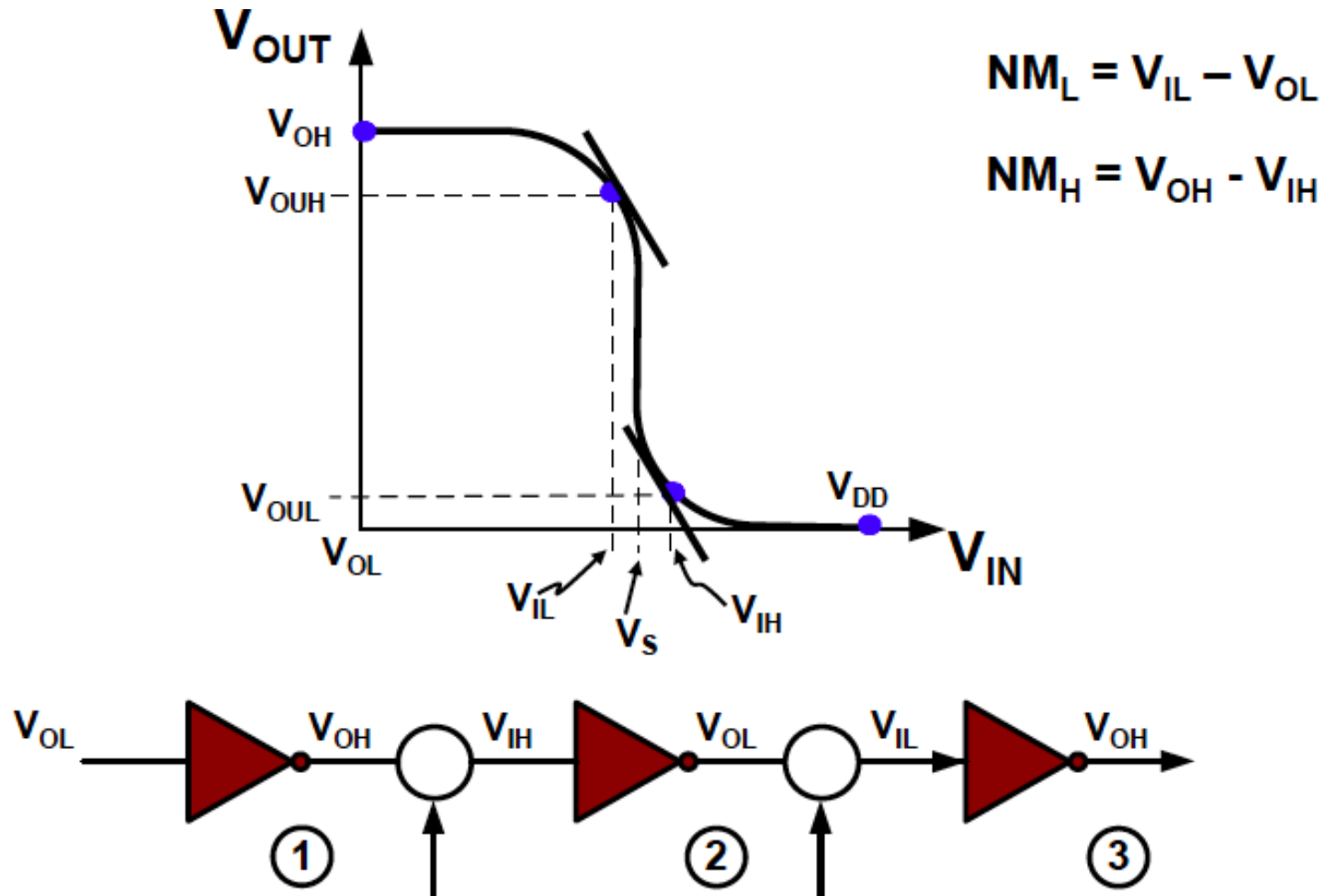
Multi-Stage Noise Margin (MSNM)

- Noise actually occurs between every gate and not at a single stage, as we have assumed so far



- In the above circuit, we need to determine how much noise we can tolerate before the circuit stops working as expected
- We note that $V_{out} = f(V_{in})$
- With noise $V_{out}' = f(V_{in}) + V_{noise} \times \text{Gain} + \text{Higher-order terms}$
- If the Gain < 1 , then noise is attenuated; otherwise it is amplified
- IDEA: Develop a metric based on keeping the Gain < 1

Classical Noise Margin



Valid Logic Gate Requirements

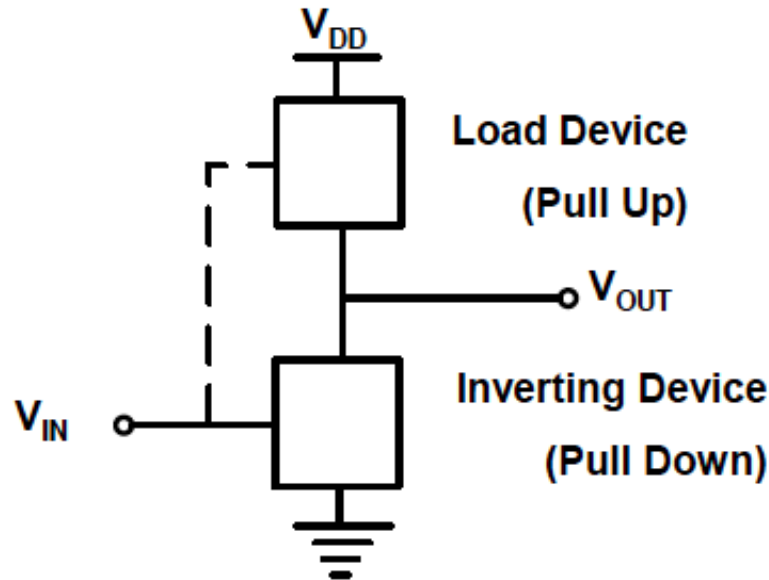
- Must have a high region between two low gain regions
- Gain must be below 1 for low gain regions
- Gain must be greater than 1 for high gain region
- Output must swing from valid low to valid high
 - Low output must be below V_{IL}
 - High output must be above V_{IH}

Noise Immunity

- Noise margin expresses the circuit ability to overpower a noise source
 - Noise sources: supply noise, cross talk, interference, offset
- Noise immunity expresses the system ability to process and transmit information correctly in the presence of noise
- For good noise immunity, the signal swing (i.e., the difference between V_{OH} and V_{OL}) and noise margin must be large enough to overpower the impact of fixed sources of noise

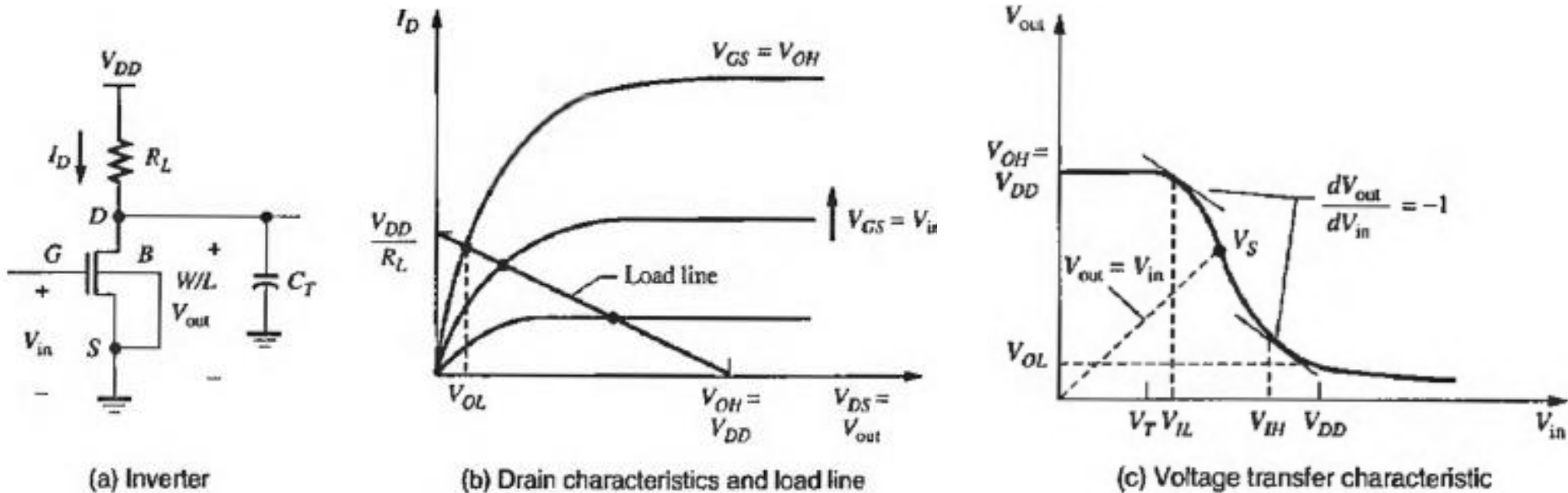
Basic MOS Inverter

- Basic structure of MOS inverter is shown below: NMOS pulldown device with a variety of possible pullup devices



- We will derive the noise margin parameters for different types of inverters

Resistive-Load Inverter Design



VTC Critical Points:

V_{OH} , V_{OL} , V_{IL} , V_{OL} , V_S

Midpoint of VTC: $V_{in} = V_{out} = V_S$

VTC Critical Points

Calculating VTC Critical Points

1. $V_{OH} = V_{DD}$: NMOSFET off

2. V_{OL} : NMOSFET on, resistive region (V_{DS} low)

Voltage divider action

$$I_R = I_{DS} \text{ (lin)}$$

$$\frac{k}{2}[2(V_{GS} - V_T)V_{DS} - V_{DS}^2] = \frac{(V_{DD} - V_{DS})}{R_L}$$

When $V_{GS} = V_{OH} = V_{DD}$, $V_{DS} = V_{OL}$

$$\frac{V_{DD} - V_{OL}}{R_L} = \frac{k}{2}[2(V_{OH} - V_T)V_{OL} - V_{OL}^2]$$

$$V_{OL}^2 - 2\left(\frac{1}{kR_L} + V_{DD} - V_T\right)V_{OL} + \frac{2V_{DD}}{kR_L} = 0$$

VTC Critical Points

For small V_{OL} , $V_{OL} \approx \frac{V_{DD}}{1 + kR_L(V_{DD} - V_T)}$

Increase k or R_L to make V_{OL} small, at expense of area;

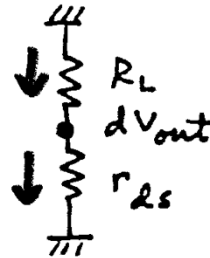
also DC power $P = IV = \left(\frac{V_{DD} - V_{OL}}{R_L} \right) V_{DD}$

Get V_{IL} , V_{IH} at $\frac{\partial V_{out}}{\partial V_{in}} = -1.0$ (Why?)

Small Signal Analysis

$$d I_D = -d V_{out} (G_L + g$$

$$\text{Or } d V_{out} = -d I_D (R_L \parallel r_d$$



$$3. V_{in} = V_{IL} \rightarrow V_{out} \approx V_{DD} \rightarrow r_{ds} \gg R_L$$

$$I_R = I_{DS} (\text{sat})$$

$$I_D = \frac{k}{2} (V_{GS} - V_T)^2 \quad (\text{sat.}) \text{ when } V_{GS} = V_{in} = V_{IL}$$

VTC Critical Points

$$\frac{V_{DD} - V_{out}}{R_L} = \frac{W\mu_n C_{ox}(V_{in} - V_T)^2}{2L} \quad \{*\}$$

Differentiating with respect to V_{in} yields

$$-\frac{1}{R_L} \left(\frac{\partial V_{out}}{\partial V_{in}} \right) = k(V_{in} - V_T)$$

Setting $V_{in} = V_{IL}$ and $\frac{\partial V_{out}}{\partial V_{in}} = -1.0$ gives

$$k(V_{IL} - V_T)R_L = 1 \quad \text{or} \quad V_{IL} = V_T + \frac{1}{kR_L}$$

V_{out} can be obtained from $\{*\}$

VTC Critical Points

4. At $V_{in} = V_{IH}$, $V_{out} \sim 0$ and MOSFET in linear region

$$k \left[(V_{in} - V_T) V_{out} - \frac{V_{out}^2}{2} \right] \approx \frac{V_{DD} - V_{out}}{R_L}$$

Differentiating with respect to V_{in} yields

$$\left[(V_{in} - V_T) \frac{\partial V_{out}}{\partial V_{in}} + V_{out} - V_{out} \frac{\partial V_{out}}{\partial V_{in}} \right] = -\frac{1}{R_L} \frac{\partial V_{out}}{\partial V_{in}}$$

Setting $V_{in} = V_{IH}$ and $\frac{\partial V_{out}}{\partial V_{in}} = -1.0$ gives

$$V_{IH} = V_T + 2V_{out} - \frac{1}{kR_L} \quad \text{and}$$

$$\frac{k}{2} [2(V_{IH} - V_T) V_{out} - V_{out}^2] = \frac{V_{DD} - V_{out}}{R_L} \quad \{**\}$$

VTC Critical Points

V_{out} can be obtained from $\{**\}$

Then, we get
$$V_{IH} = V_T + \sqrt{\frac{8V_{DD}}{3kR_L}} - \frac{1}{kR_L}$$

Alternatively,

$$\frac{dV_{out}}{dV_{in}} \approx -\frac{dI_D}{dV_{in}} r_{ds} = -\frac{dI_D}{dV_{GS}} \frac{dV_{DS}}{dI_D} = -1$$

$$I_D = \frac{k}{2}[2(V_{GS} - V_T)V_{DS} - V_{DS}^2]$$

Therefore,
$$\frac{dV_{out}}{dV_{in}} = \frac{-kV_{DS}}{k(V_{GS} - V_T - V_{DS})} = -1$$

This gives
$$V_{out} = \frac{V_{in} - V_T}{2}$$

VTC Critical Points

We also have

$$\frac{k}{2}[2(V_{in} - V_T)V_{out} - V_{out}^2] = I_D = \frac{(V_{DD} - V_{out})}{R_L}$$

This gives a quadratic equation for $V_{in} = V_{IH}$!

5. At $V_S = V_{in} = V_{out}$, $V_{GS} = V_{DS}$ (MOSFET in saturation)

$$\frac{k}{2}(V_M - V_T)^2 = \frac{(V_{DD} - V_M)}{R_L}$$

Can either solve this quadratic in V_S or iterate with initial guess to be $V_{DD}/2$

HJS Ex. 4.2

Noise Margins for R-Load Inverter

Problem:

Given the following data for an inverter like the one shown in Figure 4.9a, find the multi-source noise margins.

$$k' = \mu_n C_{ox} = 430 \mu\text{A/V}^2, \quad V_T = 0.4 \text{ V}, \quad \frac{W}{L} = 2.0$$
$$V_{DD} = 1.2 \text{ V}, \quad R_L = 20 \text{ k}\Omega$$

HJS Ex. 4.2

Solution:

$$V_{OH} = V_{DD} = 1.2 \text{ V} \quad \text{and} \quad k = k' \left(\frac{W}{L} \right) = 430 \frac{\mu\text{A}}{\text{V}^2} \times 2.$$

Find V_{OL} by solving (4.6d) with the given data (assuming transistor is linear):

$$V_{OL} \approx \frac{V_{DD}}{1 + kR_L(V_{DD} - V_T)} = \frac{1.2}{1 + (430 \times 10^{-6})(2)(20 \times 10^3)(1.2 - 0.4)} \approx 0.08 \text{ V}$$

Note that V_{OL} is safely below V_T . Also, the assumption that the transistor is in the linear region is valid.

Find V_{IL} using (4.8b):

$$V_{IL} = \frac{1}{kR_L} + V_T = 0.46 \text{ V}$$

A check confirms that the transistor is in the saturation region, as assumed.

HJS Ex. 4.2

Find V_{IH} from (4.8f):

$$V_{IH} = V_T + \sqrt{\frac{8V_{DD}}{3kR_L} - \frac{1}{kR_L}} = 0.4 + \sqrt{\frac{8}{3} \frac{1.2}{(2)430(10^{-6})20(10^3)}} - \frac{1}{(2)430(10^{-6})20(10^3)} \approx 0.77 \text{ V}$$

Again, the transistor is in the linear region, as assumed. The noise margins are then

$$NM_L = V_{IL} - V_{OL} = 0.46 - 0.08 = 0.38 \text{ V}$$

$$NM_H = V_{OH} - V_{IH} = 1.2 - 0.77 = 0.43 \text{ V}$$

HJS Ex. 4.3

SPICE Comparison to Hand Analysis in 0.18 μm resistive-load NMOS inverter

Problem:

Compare SPICE VTC results to hand calculations for a resistive-load NMOS inverter with $R_L = 30\text{ k}\Omega$, $W = 400\text{ nm}$, $L = 200\text{ nm}$ (i.e., $W/L = 2$). That is, find V_{OH} , V_{OL} , V_{IH} , V_{IL} , and V_S . Useful parameters for 0.18 μm CMOS are as follows:

$$k' = \mu_n C_{ox} = 270\text{ }\mu\text{A/V}^2, \quad C_{ox} = 1\text{ }\mu\text{F/cm}^2, \quad V_T = 0.5\text{ V}$$

$$E_C L = 1.2\text{ V}, \quad V_{DD} = 1.8\text{ V}, \quad v_{sat} = 8 \times 10^6\text{ cm/s}$$

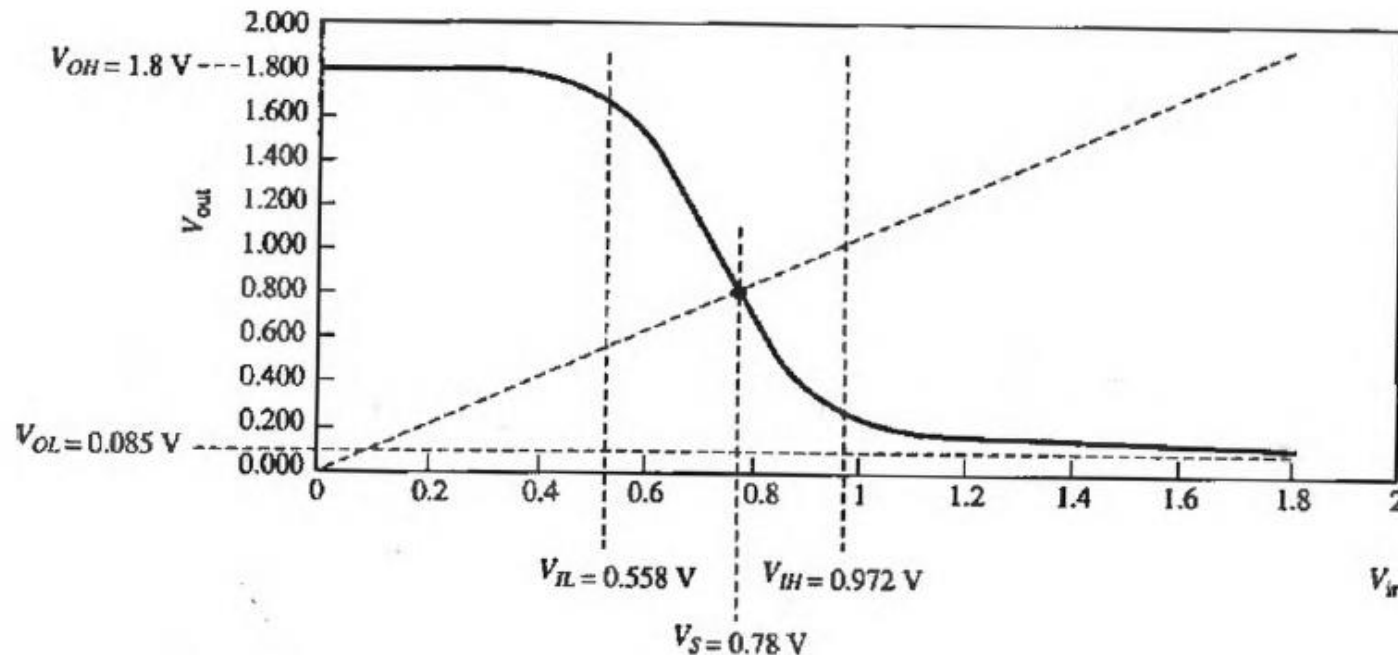
HJS Ex. 4.3

The SPICE file produces the following VTC. The graph is annotated with the key points. Using these values, the noise margins and switching threshold are

$$NM_L = V_{IL} - V_{OL} = 0.558 - 0.085 = 0.473 \text{ V}$$

$$NM_H = V_{OH} - V_{IH} = 1.8 - 0.972 = 0.828 \text{ V}$$

$$V_S = 0.78 \text{ V}$$



HJS Ex. 4.3

To compare against hand calculation, we first find $V_{OH} = V_{DD} = 1.8 \text{ V}$.

Next find V_{OL} :

$$V_{OL} \approx \frac{V_{DD}}{1 + kR_L(V_{DD} - V_T)} = \frac{1.8}{1 + (2) \left(270 \frac{\mu\text{A}}{\text{V}^2} \right) (30 \text{ k}\Omega) (1.8 - 0.5)} \approx 0.082 \text{ V}$$

Then find V_{IL} :

$$V_{IL} = \frac{1}{kR_L} + V_T = 0.06 + 0.5 = 0.56 \text{ V}$$

Find V_{IH} as follows:

$$V_{IH} = V_T + \sqrt{\frac{8V_{DD}}{3kR_L}} - \frac{1}{kR_L} = 0.5 + 0.54 - 0.06 = 0.98 \text{ V}$$

HJS Ex. 4.3

The noise margins are

$$N_{ML} = V_{IL} - V_{OL} = 0.478 \text{ V}$$

$$N_{MH} = V_{OH} - V_{IH} = 0.817 \text{ V}$$

These are quite close to the SPICE results.

The last step is to compute the switching threshold using Equation (4.9):

$$\frac{Wv_{sat}C_{ox}(V_S - V_T)^2}{(V_S - V_T) + E_C L} = \frac{V_{DD} - V_S}{R_L}$$

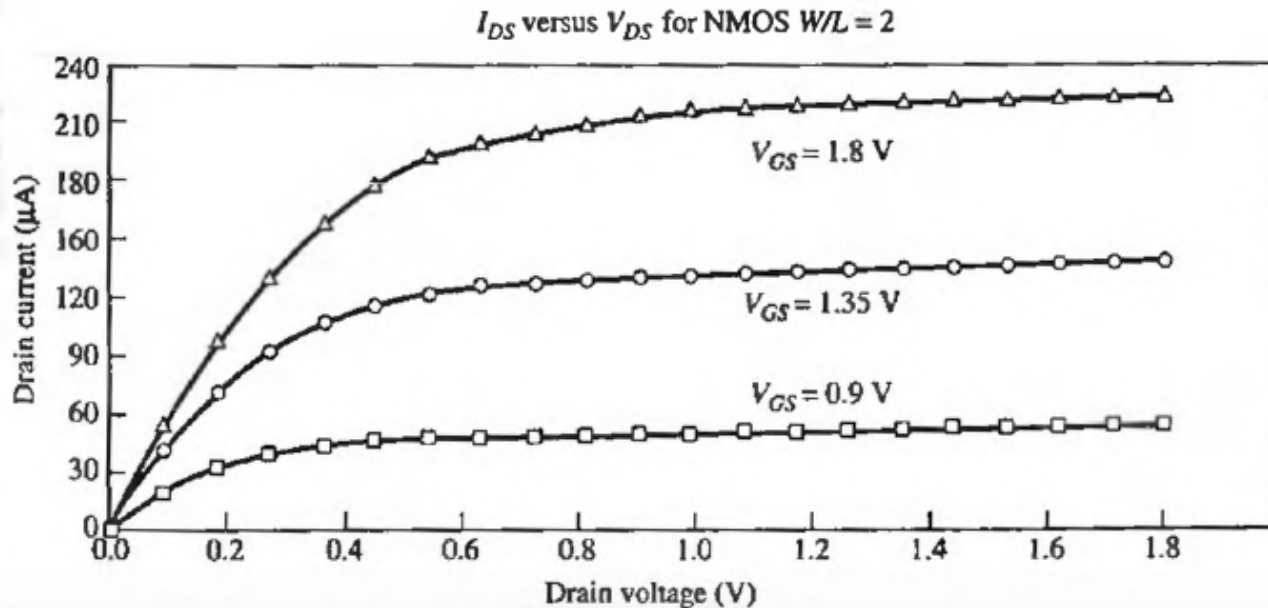
$$\frac{0.4(10^{-4})(8 \times 10^6)(10^{-6})(V_S - 0.5)^2}{(V_S - 0.5) + 1.2} = \frac{1.8 - V_S}{30 \text{ k}\Omega}$$

This equation can be rearranged and solved to produce

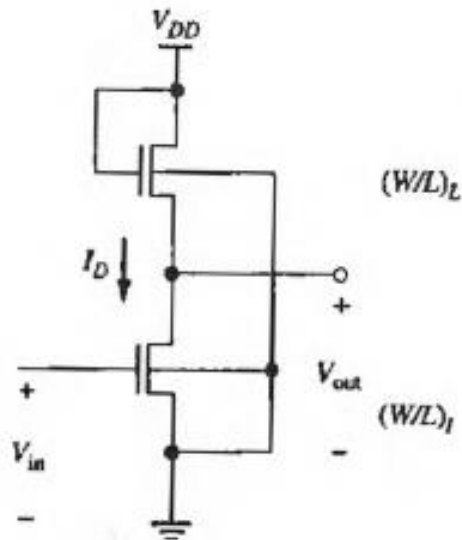
$$V_S = 0.89 \text{ V}$$

HJS Ex. 4.3

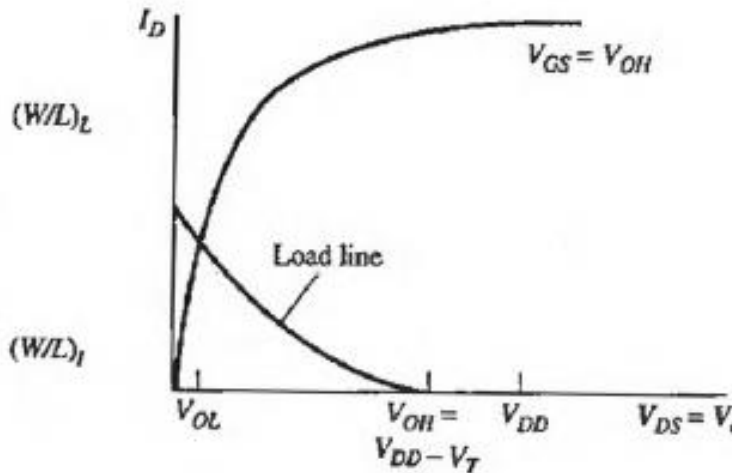
However, this value is about 15% higher than the SPICE result. Part of the reason is that the channel-length modulation parameter has been neglected in this calculation. Recall that the transistor is in saturation when computing V_S . By examining the SPICE curve shown below for $V_{GS} = 0.9$ V, there is a noticeable increase in the current in saturation due to channel-length modulation and DIBL. If we included a term $(1 + \lambda V_D)$ in the current equation and decreased V_T due to DIBL, the actual saturation current would be larger. This would reduce the switching threshold to 0.78 V rather than the value of 0.89 V computed by hand.



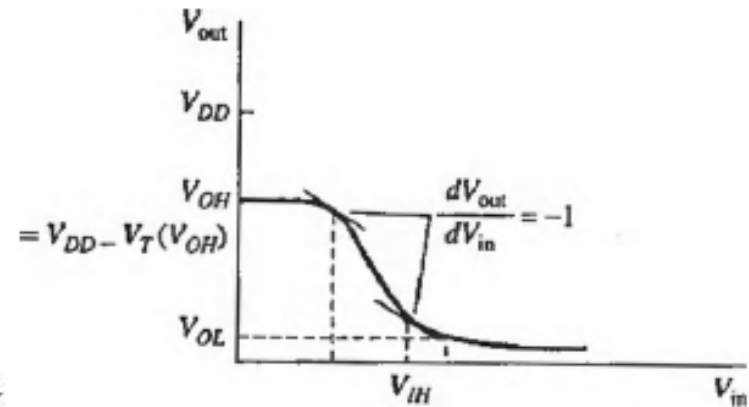
Saturated Enhancement-Load Inverter



(a) Enhancement-loaded inverter, saturated load



(b) Drain characteristic and load line



(c) Voltage transfer characteristic

- Single NMOSFET with gate and drain connected as load or pullup device
- **Drawback:** $V_{OH} \neq V_{DD}$

VTC Critical Points

For MOSFET, the **aspect ratio**, W/L , is important

$$K_R = \frac{k_{\text{invert}}}{k_{\text{load}}} = \frac{k' (W/L)_I}{k' (W/L)_L} = \frac{(W/L)_I}{(W/L)_L}$$

Increase K_R to increase V_{OL} ; like to have $V_{OL} \sim 0.05V_{DD}$

Increasing K_R increases circuit area; $[(W/L)_I(W/L)_L]^{1/2} \sim 1$

Calculating V_{OH} , V_{OL} , V_{IL} , V_{IH} and V_S :

$$1. V_{OH} = V_{DD} - V_{TL}(V_{OH})$$

$$V_{TL} = V_{TD} + \gamma(\sqrt{V_{SB} + 2|\phi_F|} - \sqrt{2|\phi_F|})$$

where $V_{SB} = V_{OH}$

If $V_{DD} = 1.2V$, $V_{TL}(V_{OH}=1.2V) = 0.45V$,

$V_{OH} = 1.2 - 0.45 = 0.75V$! **Impractical!**

VTC Critical Points

2. Find V_{OL} : Load transistor always saturates ($V_{GS} = V_{DS}$)

$$I_{DI}(\text{lin}) = I_{DL}(\text{sat})$$

$$\frac{W_I}{L_I} \frac{\mu_n C_{ox}}{\left(1 + \frac{V_{out}}{E_{CN} L_I}\right)} \left[(V_{in} - V_{TL}) V_{out} - \frac{V_{out}^2}{2} \right] = \frac{W_L \mu_{sat} C_{ox} (V_{DD} - V_{out} - V_{TL})^2}{(V_{DD} - V_{out} - V_{TL}) + E_{CN} L_L}$$

Tedious to calculate V_{OL} by hand.

- Major drawback of this inverter is significant DC power loss when the output is low.

HJS Ex. 4.4

Computation of V_{OH} for a Saturated Enhancement Load Inverter

Problem:

Calculate the V_{OH} of the NMOS inverter shown in Figure 4.10a. Assume the following parameters for a $0.13\ \mu\text{m}$ technology. Comment on the magnitude of the result:

$$2|\phi_F| = 0.88\ \text{V}, \quad \gamma = 0.2\ \text{V}^{1/2}, \quad V_T = 0.4\ \text{V}, \quad V_{DD} = 1.2\ \text{V}$$

HJS Ex. 4.4

Solution:

$$\begin{aligned}V_{OH} &= V_{DD} - V_{TL} = V_{DD} - V_{TL}(V_{SB}) \\&= V_{DD} - [V_{T0} + \gamma(\sqrt{V_{SB} + 2|\phi_F|} - \sqrt{2|\phi_F|})] \\&= 1.2 - [0.4 + 0.2(\sqrt{V_{OH} + 2|-0.44|} - \sqrt{2|-0.44|})] \\&= 0.8 - 0.2\sqrt{V_{OH} + 0.88} + 0.2\sqrt{0.88} \\&= 0.99 - 0.2\sqrt{V_{OH} + 0.88}\end{aligned}$$

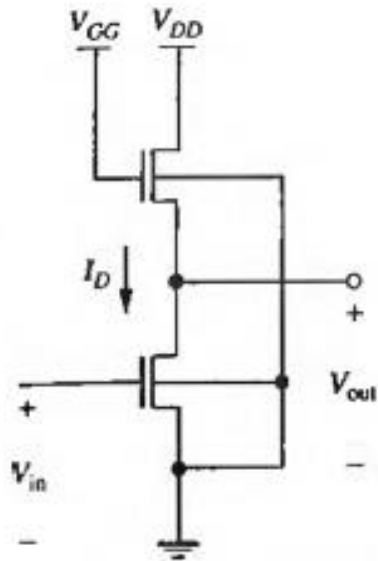
Solve by iteration, starting at the (optimistic) value $V_{OH} = 1.2$ V.

Old V_{OH}	New V_{OH}
1.20	0.70
0.70	0.74
0.74	0.73
0.73	0.73

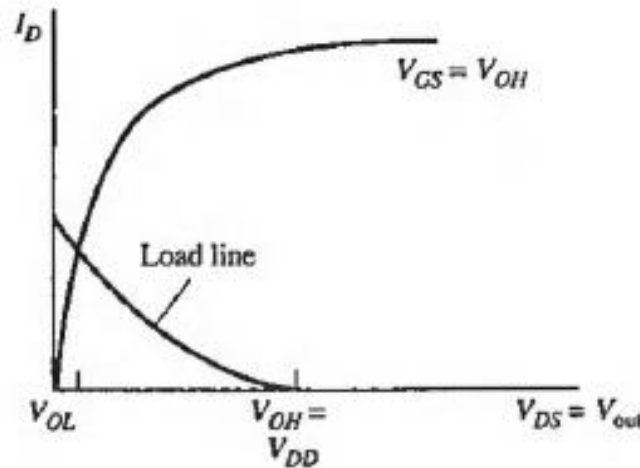
$$\therefore V_{OH} = 0.73 \text{ V}$$

Clearly, a V_{OH} of 0.73 V from a 1.2 V supply is not acceptable as input to the next gate. Also, note that faster convergence would have been obtained had we started with $V_{OH} = 1.2 - 0.4 = 0.8$ V.

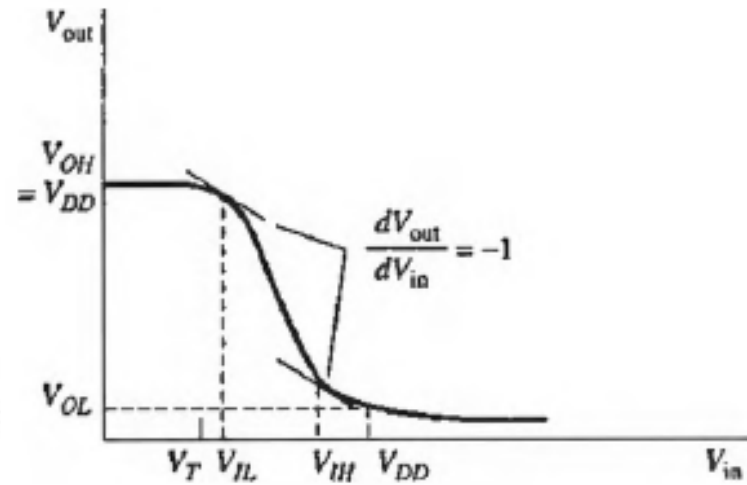
Linear Enhancement-Load Inverter



(a) Enhancement-loaded inverter, linear load



(b) Drain characteristic and load line



(c) Voltage transfer characteristic

- Single NMOSFET with gate voltage above V_{DD} as load or pullup device

Linear Enhancement-Load Inverter

- Connect the gate of load NMOSFET to dc voltage supply, V_{GG} , instead of to V_{DD} , to increase V_{OH} .

Desired gate voltage is

$$V_{GG} > V_{DD} + V_{TL}(V_{DD})$$

- More chip area is used because
 - Extra voltage source V_{GG}
 - Required K_R value higher than saturated enh. load
 - DC power dissipation in output low state
- Hence, rarely used

HJS Ex. 4.5

Design of Saturated Load Inverter

Problem:

Design the saturated enhancement load inverter of Figure 4.10a such that it delivers a low output voltage of $V_{OL} = 0.1$ V when the input is V_{DD} . Assume a $0.13\text{ }\mu\text{m}$ technology, with $L = 100$ nm and let the pull-up device be of unit size. Use the parameters

$$\mu_n = 270\text{ cm}^2/\text{V}\cdot\text{s}, \quad C_{ox} = 1.6\text{ }\mu\text{F}/\text{cm}^2, \quad V_{T0} = 0.4\text{ V}, \quad V_{DD} = 1.2\text{ V}$$

$$E_{CNL} = 0.6\text{ V}, \quad v_{sat} = 8 \times 10^6\text{ cm/s}, \quad \gamma = 0.2\text{ V}^{1/2}$$

HJS Ex. 4.5

Solution:

Use the current equation for the pull-down in the linear region and the pull-up in the saturation region.

$$\frac{W_I}{L_I} \frac{\mu_n C_{ox}}{\left(1 + \frac{V_{out}}{E_{CN} L_I}\right)} \left[(V_{in} - V_{TH}) V_{out} - \frac{V_{out}^2}{2} \right] = \frac{W_L \mu_{sat} C_{ox} (V_{DD} - V_{out} - V_{TL})^2}{(V_{DD} - V_{out} - V_{TL}) + E_{CN} L_L}$$

We can eliminate C_{ox} and then set $V_{in} = V_{DD}$ (we should really set it to V_{OH} but we keep things simple in this problem) and $V_{out} = V_{OL}$:

$$\therefore \frac{W_I}{L_I} \frac{\mu_n}{\left(1 + \frac{V_{OL}}{E_{CN} L_I}\right)} \left[(V_{DD} - V_{TH}) V_{OL} - \frac{V_{OL}^2}{2} \right] = \frac{W_L \mu_{sat} (V_{DD} - V_{OL} - V_{TL})^2}{(V_{DD} - V_{OL} - V_{TL}) + E_{CN} L_L}$$

Since we know that the output voltage is $V_{OL} = 0.1$ V, we can compute the correct V_{TL} :

$$\begin{aligned} V_T &= [V_{T0} + \gamma(\sqrt{V_{SB} + 2|\phi_F|} - \sqrt{2|\phi_F|})] \\ &= [0.4 + 0.2(\sqrt{0.1 + 2|-0.44|} - \sqrt{2|-0.44|})] \\ &= 0.4 + 0.2(\sqrt{0.1 + 0.88} - \sqrt{0.88}) \\ &= 0.41 \text{ V} \end{aligned}$$

HJS Ex. 4.5

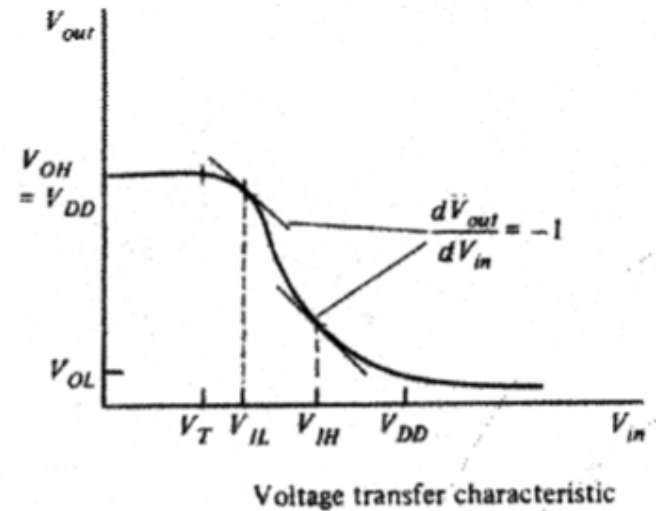
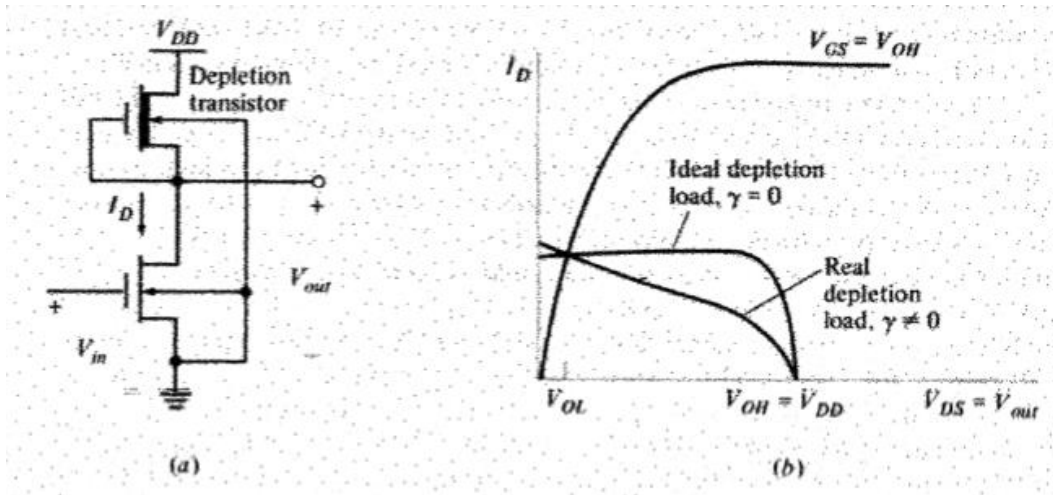
The degree of body effect is small, as one would expect with a body-bias of 0.1 V. We can substitute the known quantities and solve for the transistor sizes:

$$\begin{aligned}\therefore \frac{W_I}{0.1(10^{-4})\text{cm}} \frac{(270 \text{ cm}^2/\text{V}\cdot\text{s})}{(1 + (0.1/0.6))} \left[(1.2 - 0.4)0.1 - \frac{0.1^2}{2} \right] \\ = \frac{W_L(8 \times 10^6 \text{ cm/s})(1.2 - 0.1 - 0.41)^2}{(1.2 - 0.1 - 0.41) + 0.6}\end{aligned}$$

$$\therefore K_R = \frac{W_I}{W_L} = \frac{2.95}{1.75} = 1.7$$

If $W_L = 100 \text{ nm}$, then $W_I = 170 \text{ nm}$.

Depletion Load Inverter



- Single depletion-mode NMOSFET as load or pullup device

Depletion-Load Inverter

- Connect the gate of depletion-mode, load NMOSFET output node, without external gate drive and output level to V_{DD}
- Advantages and Disadvantages
 - Small chip area due to no extra dc voltage source and smaller required K_R value than enhancement loads
 - DC power dissipation in output low state
- Most difficulty is the control of threshold voltage of depletion-mode transistor in manufacturing
- Not used now